Lesson plan · for the teacher

Logic: switches that decide

IoT · Level 2 · Lesson 2

Time
40 min

Materials
Two cards per pupil, one marked 1 and one marked 0; Chalk or markers for truth tables; Optional: two switches (or paper-clip switches), two AA batteries, an LED and a 100 Ω resistor

Learning goals

- Describe AND, OR and NOT with switches and truth tables.
- Build AND and OR from two switches, in series and in parallel, with kit or on paper.
- Write a classroom rule (a fan, an alarm, a stair light) as logic.

Activity: Human logic gates

1. Warm-up (5 min): Ask: “When should our fan run, and when is it wasting power?” Write pupils' rules on the board in their own words.
2. Teach (10 min): Draw two switches in series (AND) and in parallel (OR), each with a battery and a lamp. Add the fridge light as NOT. Write 1 for on and 0 for off.
3. Act it out (12 min): Two pupils hold up 1 or 0 cards; a third is the gate and stands for 1 or sits for 0. Run all four rows for AND, then OR, then NOT with one input. The class checks each answer.
4. Tables (8 min): Groups write “hot AND occupied” as a four-row table, then mark the row that saves power in an empty room.
5. Wrap-up (5 min): Each group shares one school rule written with AND, OR or NOT. Exit ticket: the fan rule and its truth table.

Prepare 1 and 0 cards in advance; scrap paper is fine. If you have kit, wire the AND and OR circuits before class and test them.

Discussion

- Where do you see AND and OR rules at school or at home?
- Why write a truth table before building anything?
- Which classroom rule would you add NOT to, and why?

Exit ticket

- Write the fan rule with AND, and fill in its four-row truth table.

Try it at school tomorrow
Human logic gates: Give pairs of pupils cards marked 1 and 0, and let a third pupil be the gate: standing for 1, sitting for 0. Run AND, OR and NOT for every row, then have groups write the classroom fan rule as a truth table. If you have kit, build AND and OR with two switches, two AA batteries, an LED and a 100 Ω resistor.

Practice questions · for you

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1 **A water-tank alarm beeps if the tank is nearly empty OR the pump is overheating. Today the tank is full, but the pump is overheating. What happens?**

- ◯ It beeps: one input on is enough for OR
- ◯ It beeps only if the tank is also nearly empty
- ◯ It stays silent, because the tank is full

2 **Two switches wired one after the other, in series, make an OR: closing either one lights the lamp.**

- ◯ True
- ◯ False

3 **Match each rule to its gate.**

Fan: hot, and someone is in the room	• • • •	OR
Alarm: front door or back door open	• • • •	XOR
Fridge light: on when its switch is not pressed	• • • •	AND
Stair light: on when the two switches differ	• • • •	NOT

4 **Complete the sentence about truth tables.**

A rule with two inputs has _____ rows in its truth table; a rule with three inputs has _____.

Word bank: 6 · 4 · 3 · 8

5 **It's 2 pm on a hot day, and the room is empty (hot = 1, occupied = 0). Which rules would switch the fan on?**

- ☐ hot AND NOT occupied
- ☐ hot OR occupied
- ☐ hot AND occupied
- ☐ NOT hot OR occupied

✎ **Which everyday rule at your school could you write with AND, OR or NOT? Write it as a sentence your pupils would understand.**

Check your answers

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1 **It beeps: one input on is enough for OR**

0 OR 1 is 1: one problem is enough to sound an OR alarm.

2 **False**

In series, current must pass through both switches, so both must be closed: that's AND.

3 **Fan: hot, and someone is in the room — AND**
Alarm: front door or back door open — OR
Fridge light: on when its switch is not pressed — NOT
Stair light: on when the two switches differ — XOR

Listen for the words: "and" means all must be true, "or" means any one is enough, "not" flips it, and "differ" is XOR.

4 **A rule with two inputs has 4 rows in its truth table; a rule with three inputs has 8.**

Each input can be 0 or 1, so every extra input doubles the rows: 2 × 2 = 4, and 2 × 2 × 2 = 8.

5 **hot OR occupied; hot AND NOT occupied**

1 OR 0 = 1, and 1 AND NOT 0 = 1 AND 1 = 1, so those two switch it on.